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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT		
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Project Title

Adaptive AI Research Intelligence Platform for Automated Literature Review and Research Gap Discovery

Abstract

The rapid growth of scientific publications across various domains has made it increasingly difficult for researchers and students to efficiently identify relevant literature, analyze research trends, and discover potential research gaps. Existing research tools primarily focus on keyword-based search and paper retrieval, requiring significant manual effort to review large volumes of literature. This creates challenges in conducting comprehensive literature reviews and identifying emerging research opportunities.

The proposed **Adaptive AI Research Intelligence Platform** addresses this problem by providing an intelligent system that automates literature analysis, summarizes research papers, identifies related studies, and discovers potential research gaps. The platform adapts to user interests and continuously updates recommendations based on newly published research and user interactions. It also assists researchers in understanding research trends, citation patterns, and unexplored areas within a domain.

Research Gap: Current literature analysis systems mainly provide search and summarization capabilities but lack adaptive mechanisms for personalized research assistance, automated research gap identification, trend analysis, and intelligent recommendation of future research directions. Most existing solutions require significant manual intervention and do not continuously adapt to changing research landscapes.

Software and Technologies Used: React.js, Spring Boot, PostgreSQL, REST APIs, Docker, Kubernetes, GitHub Actions, Argo CD, LangChain, Vector Database (ChromaDB), and OpenAI-based Large Language Models.

Real-Life Impact:

- Reduces time spent on literature review.
- Helps identify novel research opportunities.
- Improves research productivity and decision-making.
- Supports adaptive and personalized research assistance.
- Enables efficient knowledge discovery from large-scale scientific literature.

Working Flow:

User Login



Paper Upload / Paper Retrieval



Data Extraction & Processing



AI Literature Analysis



Paper Classification & Summarization



Adaptive Recommendation Engine



Research Gap Discovery



Trend & Citation Analysis



Interactive Dashboard



User Feedback



Continuous System Adaptation

Expected Outcome: The proposed system will enable researchers to perform faster and more effective literature reviews, automatically identify research gaps, generate concise summaries, discover emerging research trends, and receive adaptive recommendations for future research directions. The platform aims to improve research productivity, reduce manual effort, and support data-driven scientific knowledge discovery through adaptive software engineering principles.

Signature of the Guide

HOD